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COMMENTARY

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The potential of neuroarchitecture and 4E-Cognition: From microbial dynamics to active environments and back via scalable experimental designs (commentary on Wang et al., 2024)

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1 | INTRODUCTION

Wang et al.'s (2024) article is relevant due to several reasons. The obvious is previous work's replication and expansion (Djebbara et al., 2019, 2021) by exploring how different forms of movement (walking/keyboard press) and environments (2D/3D) affect affordance perception. *Replication* and *expansion* of experimental effects using diverse populations and contexts is highly relevant to the still fledgling fields of *mobile brain/body imaging* (MoBI) and *neuroarchitecture*; it should be facilitated and encouraged. Furthermore, their study implements a scalable experimental design (SED; Parada, 2018) across multiple articles (Djebbara et al., 2019, 2021; Wang et al., 2024), which is particularly relevant for applied neuroscience.

In addition to providing insights into how natural, built, digital, and symbolic (NaBDS) environments impact cognition from the perspective of 4E-Cognition,[1] this commentary also seeks to bridge the gap between MoBI and the *microbiome of the built environment* (MoBE), which is a relevant aspect of 4E-Cognition (Palacios-García et al., 2022). By furthering the integration of these two frameworks (Palacios-García et al., 2022; Palacios-García & Parada, 2021), we aim to propose actionable steps that merge the physiological and microbial aspects of human–environment interactions. This integration emphasizes the need for evidence-based design features that promote both cognitive and microbial health in NaBDS environments.

Even though we promote the 3E-Cognition[2] principles for applied neuroscience (e.g. neuroarchitecture; Parada et al., 2024), here we will nevertheless contextualize Wang et al.'s findings within the broader perspective offered by 4E-Cognition (Figure 1). By discussing how

Abbreviations: MoBI, mobile brain/body imaging; SEDs, scalable experimental designs; NaBDS, natural, built, digital, and symbolic environments; MoBE, microbiome of the built environment; 4E, embodied, embedded, extended, enactive; 3E, embodied, environmentally scaffolded, enactive.

[1]Embodied, embedded, extended, and enacted (for details, see Grasso-Cladera et al., 2023)

[2]Embodied, environmentally scaffolded, and enactive.

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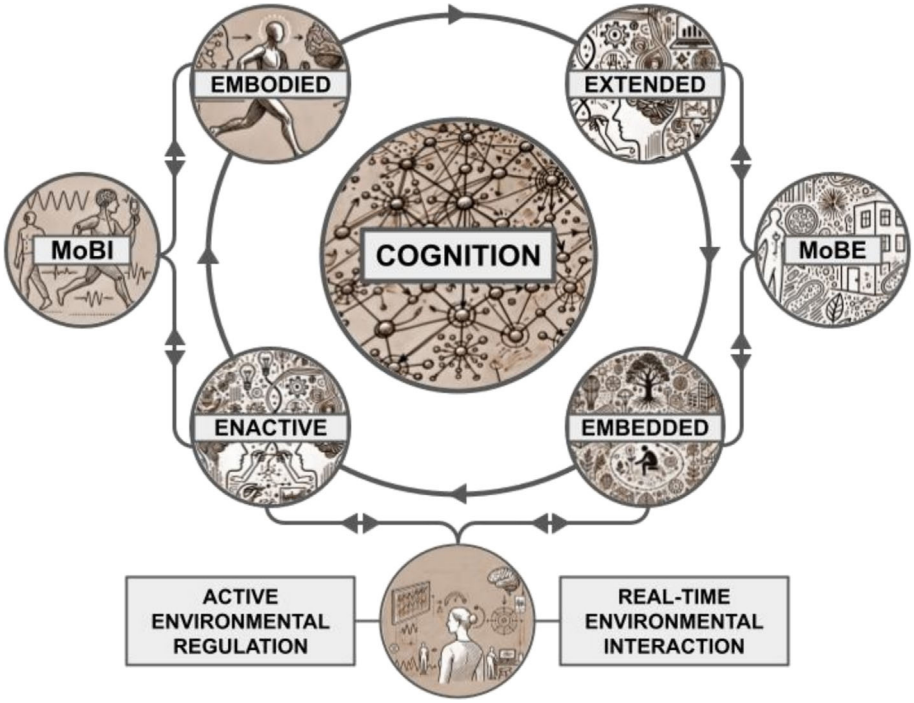

FIGURE 1 Graphical representation of the key connections between 4E-Cognition (embodied, embedded, extended, enactive), mobile brain/body imaging (MoBI) and the microbiome of the built environment (MoBE). Embodied and enactive principles, focusing on the role of the organism and active engagement with the environment, are explored through MoBI. Embedded and extended principles, highlighting environmental influence and the extension of cognitive processes, are studied via MoBE. Active environment regulation and real-time environmental interactions are applications derived from enactive and embeddedness principles. Multidirectional arrows indicate potential applications across all 4E principles.

MoBI and MoBE can converge, we present a holistic approach to understanding and enhancing human–environment interactions. This sets the stage for the subsequent sections, where each 4E principle will guide our exploration of the implications and potential applications of Djebbara, Gramann, and more recently, Wang's research.

1.1 | Embodiment: cognition unfolds within organisms

In a nutshell, *embodiment* claims cognition emerges from an organism's brain/body morpho-physiological dynamics (Grasso-Cladera et al., 2023). Wang et al. (2024) examine how physical walking versus keyboard movements influence the perception of affordances, studying the role of bodily dynamics in cognition. Findings indicate affordance perception starts early as sensorimotor automated processes, later transitioning to being influenced by movements, context, etc.

The results show that considering both *the gist*—automatic perceptual processes—and *the deep dive*—bodily action in the world—in architectural design is key. Hence, for neuroarchitecture and design, this principle implies that spaces should be crafted to facilitate and promote bodily action, enhancing the intuitive perception of affordances and promoting more effective interactions with the environment. A concrete example could be the design of workspaces that encourage natural movement throughout the day, such as standing desks that allow for easy transition between sitting and standing, or open office layouts that require walking to access shared resources. These designs not only promote physical health but also enhance cognitive engagement by aligning bodily movement with the intuitive perception of affordances within the space.

1.2 | Embeddedness: rooted in the world

Briefly, *embeddedness* claims organisms' brain/body morpho-physiological dynamics are *always* situated within specific NaBDS environments and contexts. Hence, cognition is a relational brain/body-world coupling property (Rojas-Líbano & Parada, 2019). Wang et al. focused on expanding Djebbara et al.'s work (2019, 2021); their results directly support the Embeddedness principle.

For architects, urban planners and policy-makers, this means that the design of spaces must take into account how these will be perceived and interacted with by users. These users are not homogeneous theoretical beings; they are complex individuals with history, biases, preferences and more! Co-designing with users is necessary as well as incorporating features that naturally align with human perceptual, and cognitive processes can lead to the design

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of physiology-friendly, intuitive and accessible spaces that not only hold or facilitate cognition, but might also improve it (Parada et al., 2024).

1.3 | Extendedness: unbounded by boundaries

Simply put, the *extendedness* principle claims that, in time, cognition vines out beyond organisms to include aspects, objects and concepts present in NaBDS environments. By comparing neurobehavioral dynamics associated with 2D/3D environments, the target article provides the evidence of NaBDS environments' role in cognition. Their results suggest that the spatial properties of an environment can significantly alter how individuals perceive and interact with affordances, highlighting the role of environmental context in cognition.

For MoBI research in general, and specifically for studying architectural affordances using the SED heuristic, comparing mobile paradigms (Djebbara et al., 2019, 2021) with static paradigms (Wang et al., 2024) provides stronger evidence for the influence of environmental context on cognitive processes. By incorporating mobile paradigms, researchers can capture more ecologically valid data that reflects the dynamic interactions between individuals and their environments in real time.

Furthermore, this approach allows scientists to explore complex phenomena beyond architectural affordances or navigation. For instance, scientists can investigate how cognitive processes extend to interactions with digital interfaces, wearable technologies and smart environments that adapt to users' needs in real time. This could include studies on how people use augmented and mixed reality tools in everyday life, how smart home devices influence cognitive load and decision-making or how virtual collaboration tools impact team dynamics and shared cognition. These insights are crucial not only to accommodate cognitive processes but also to actively enhance them, whether through the design of intuitive digital tools, adaptive workspaces or therapeutic environments that support mental health.

1.4 | Enaction: crafting the world as it crafts us

Succinctly, *enaction* emphasizes the historical nature of life and mind as ongoing brain/body-world couplings co-define the organism and its world (Varela et al., 1991). That is, cognition is not merely about processing information but about actively engaging with and co-creating our environments. From such dynamic interactions between organisms and environments, the backbone of cognition—sense-making acts—emerges as the *open transition between microworlds of activity*.[3] Wang et al. offer evidence on how such transition occurs in affordance perception; from initial automated processes (<200 ms) to later context-dependent ones (>200 ms), influenced by movement type and context.

This finding and its enactive interpretation has significant implications for neuroarchitecture and design, suggesting that spaces should be designed to foster dynamic, reciprocal interactions between/within users and their environments. Unlike embodiment, which focuses on facilitating bodily action, *enaction* emphasizes the need for environments that adapt and respond to users, creating feedback loops that enhance physiological, cognitive and psycho-affective processes. For example, consider the design of interactive public spaces equipped with responsive technologies, such as sensors and smart surfaces that change in response to movement (Anghel et al., 2019; Belapurkar et al., 2018), lighting that adjusts to the presence and activities of users or pathways that subtly guide foot traffic to optimize social interactions (Magielse & Ross, 2011; Offermans et al., 2014; Shahzad et al., 2016). Such spaces not only encourage movement but also create an engaging, ever-changing environment that actively involves users in shaping their experiences.

[3]I.e., the dynamic and fluid nature of cognition as it moves through different contexts or 'microworlds'. These microworlds represent distinct, yet interconnected, domains of activity or interaction that an organism engages with as it navigates and lives its environment. These interactions are not isolated; rather, they form a continuous flow where the cognitive system transitions from one *microworld of activity* to another. For example, when a person moves from navigating a busy street (a typical public and urban microworld) to entering a quiet room (a different, private microworld), their physiological and cognitive states, sensory processes, and bodily engagements shift accordingly. These shifts are the 'open transitions', where (neuro)physiological dynamics reconfigure themselves to make sense of the new context while maintaining a connection to the previous one. These transitions are 'open' because they are not predetermined or rigidly structured; they allow for adaptation, flexibility and the incorporation of new information or experiences. This openness is central to the enactive view of cognition, which sees the mind as inherently historical and evolving, shaped by past interactions and ready to engage with future ones. See Varela et al. (1991) and Di Paolo (2018) for deeper understanding on the concept.

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2 | NEUROARCHITECTURE: 4E-COGNITION, MOBI, AND ETHICAL CONSIDERATIONS

2.1 | MoBI scalable experimental paradigms for applied neuroscience

SEDs, integrating results across multiple experiments, organically link traditional cognitive electrophysiology research and MoBI in applied neuroscience. First of all, escalating previous research to MoBI paradigms following the SED heuristic is a promising approach. Furthermore, MoBI research can become stronger by, as Wang and collaborators did, escalating paradigms back to stationary settings (Costa-Cordella et al., 2024). Thus, incorporating the SED heuristic in applied neuroscience provides robust and more comprehensive insights into how embeddedness and extendedness affect neurobehavioral dynamics.

2.2 | Active environment regulation via real-time interaction and brain/body feedback

Using real-time neurofeedback in NaBDS environments can enhance our understanding of how spaces can be designed to optimize cognitive performance and well-being. Unobtrusive MoBI technology will soon allow continuous longitudinal monitoring of brain activity while individuals navigate through spaces and perform daily activities (Wascher et al., 2023). Thus, the creation of adaptive personalized and supportive environments, actively responding to neurobehavioral, cognitive and psycho-affective states of users, might become a reality.

2.3 | MoBI meets MoBE

The *Holobiont* concept integrates the idea that cognition and behavior are influenced not only by brain/body-world couplings but also by body/environment microbial dynamics (see Palacios-García & Parada, 2021). Thus, the biogenic nature of the 4E-Cognition perspective adds a whole new layer to neuroarchitectural and affordance research allowing *MoBI to meet MoBE* research by:

1. *Incorporating evidence-based materials and design features that promote healthier MoBE in workplaces, schools, hospitals and more*. This could imply using natural, non-toxic building materials, implementing ventilation systems that reduce harmful microbial growth and incorporating indoor plants supporting beneficial microbial communities. Hospitals, healthcare and rehabilitation centers can be optimized to support beneficial microbial interactions, aiding recovery and improving health outcomes. This can include therapeutic gardens and access to natural outdoor environments for patients. Likewise, classrooms and offices can incorporate natural elements and biophilic design principles that support individual or group healthy microbial conditions, both reducing the transmission of infectious diseases and potentially improving concentration, creativity and productivity.
2. *Regulating certain features of active environments by considering specific microbial needs of individuals*. This includes tailored conditions such as humidity, temperature, and air quality to support the optimal comfort, individual health, and cognitive performance.
3. *Creating public spaces that promote interaction with diverse microbial atmospheres*. Parks, community gardens, and green/blue spaces, can be designed to encourage microbial diversity, benefiting the mental and physical health of urban populations.
4. *Influencing public health policies to consider the impact of environmental designs on health: microbial, mental, physical and/or societal*. Policies can promote the development of spaces supporting microbiomes, encouraging practices that enhance global health.

2.4 | Ethical implications

The integration of MoBI with MoBE presents exciting opportunities for advancing neuroarchitecture. However, these advancements also bring significant ethical considerations that must be addressed. As we propose new ways to design environments that respond to and support (neuro)physiological, cognitive, and microbial health, we must ensure that these implementations are equitable and accessible to all.

Historically, the built environment has perpetuated social, economic, health and other inequalities within communities and society at large—examples include redlining, designed injustice and hostile architecture. As researchers working in Santiago de Chile, we know these problems firsthand. Such practices have disproportionately affected marginalized groups, leading to long-standing disparities in access to healthy and supportive environments. As neuroarchitecture research evolves, it is imperative that we proactively acknowledge this context and commit to designing studies and interpreting results with a focus on inclusivity and social justice.

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This means that when implementing MoBI and MoBE principles, researchers and practitioners must consider how these technologies and designs can be applied in ways that do not exacerbate existing inequalities. Instead, the goal should be to use these innovations to address and rectify disparities, ensuring that all individuals and communities can benefit from healthier, more supportive environments. This requires careful consideration of who has access to these technologies, how environments are designed and the potential long-term impacts on different populations. This type of projects is already happening on the microbiology sphere (Ishaq et al., 2021); MoBI should follow suit.

Ultimately, the ethical implications of 'MoBI meets MoBE' work extend beyond privacy, consent and the manipulation of cognitive and psycho-affective phenomena. They also encompass the responsibility to create environments that are equitable and accessible, fostering well-being for all members of society without reproducing the ghosts of architectural and scientific colonialism. As researchers, designers and policy-makers, we must engage in ongoing dialogue and collaboration with diverse communities to ensure that our work contributes to a more just and inclusive future.

3 | CONCLUDING REMARKS

The target article provides evidence about the neurobehavioral dynamics of the perception of architectural affordances and the power of experimental replication and expansion by exploring how different forms of movement and environments affect affordance perception. Furthermore, we have argued that by integrating 4E principles, future research can deepen our understanding of the complex interactions between brain, body, conspecifics, other species and environment. Incorporating advanced technologies, such as MoBI, real-time feedback and MoBE, while addressing ethical considerations, will drive the field of neuroarchitecture forward, ultimately leading to the design of more supportive and adaptive environments.

AUTHOR CONTRIBUTIONS

Francisco J. Parada conceptualized and wrote the initial manuscript. Alejandra Rossi discussed the concepts and arguments, edited and wrote the final version of the manuscript. Francisco J. Parada drew eight independent figures using Dall-E (powered by OpenAI's language model, GPT-3.5; http://openai.com) and proceeded to design Figure 1 using Pixlr photo editor (https://pixlr.com/editor/). The authors acknowledge the use of ChatGPT (powered by OpenAI's language model, GPT-3.5; http://openai.com) for improving the grammatical and language accuracy of the submitted manuscript. Selected paragraphs were submitted to ChatGPT with the prompt 'Consider the following text ['TEXT']. Shorten it without losing information, do any modifications necessary to improve grammatical clarity and readability'. The results were copied into the manuscript body and edited/adjusted to fit the authors' writing style. Sometimes, ChatGPT would not make relevant changes and these were discarded.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ETHICS APPROVAL STATEMENT

No ethical approval was needed for developing the present work.

DATA AVAILABILITY STATEMENT

Considering this is a theoretical article, only bibliographic data compiled in a non-systematic way was used. All bibliographic material used is properly listed in the References section.

PEER REVIEW

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